

Risk-managed momentum in Indian equities

A point-in-time, executable test of volatility-scaled NIFTY 500 winners

Executive conclusion

Volatility scaling worked as a **risk-control mechanism**: VM cut holdout volatility to 15.08% and maximum drawdown to -15.81%, versus 21.05% and -24.94% for unscaled MOM. It did not deliver superior holdout returns or Sharpe. Fixed-allocation controls earned higher CAGR and Sharpe than VM. The evidence supports dynamic de-risking for drawdown control, but not a claim of superior timing alpha.

Implementation arithmetic

Holdout | annualized | matched executable paths

Account	Gross turnover	Effective RT cost	Cost + tax hurdle	Gross annual edge	Clears?
MOM	5.34x	50.4 bps	4.09%	28.80%	Yes
VM	4.73x	47.2 bps	3.54%	19.59%	Yes
FIX	4.92x	45.9 bps	3.51%	21.31%	Yes
FIXVOL	5.12x	46.4 bps	3.66%	22.10%	Yes

Effective RT cost = implementation-cost drag / round-trip-equivalent turnover. Gross edge reverses recorded costs and tax effects.

Holdout scorecard

Post-tax headline results with explicit pre-tax CAGR

Account	Final NAV	Post-tax CAGR	Pre-tax CAGR	Volatility	Sharpe	Max DD
MOM	INR 38.23m	24.71%	27.45%	21.05%	1.17	-24.94%
VM	INR 23.17m	16.05%	18.48%	15.08%	1.08	-15.81%
FIX	INR 27.68m	17.80%	20.18%	16.09%	1.12	-19.22%
FIXVOL	INR 28.77m	18.45%	20.92%	16.82%	1.11	-20.05%

Benchmark: NIFTY 500 TRI CAGR 13.23%. Final NAVs are not directly comparable because each account continues from a different frozen 31 March 2023 state; CAGR and normalized wealth are the appropriate comparisons.

What was tested

- Long-only, unlevered Barroso-Santa-Clara adaptation.
- Top-decile NIFTY 500 winners; monthly 11-month t-12 to t-2 signal.
- VM target: min(1, 12% / estimated volatility); residual capital in zero-yield cash.
- One execution, tax, settlement and corporate-action engine for all variants.

Why the result matters

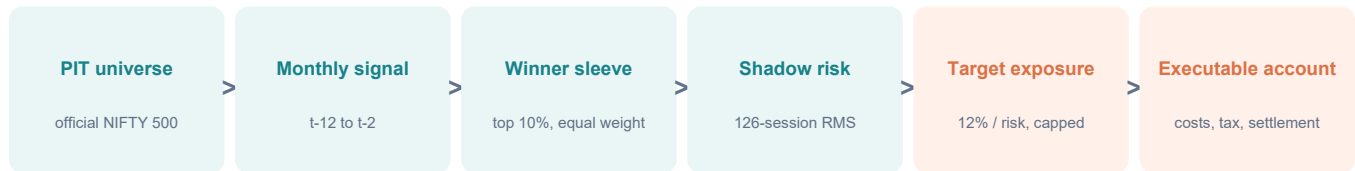
- MOM produced the strongest growth, with the largest drawdown and turnover.
- VM lowered risk partly by carrying cash when trailing risk was high.
- FIX and FIXVOL beat VM on CAGR and Sharpe, weakening the timing-alpha claim.
- The result supports risk control, not return enhancement.

Strategy variants

MOM is the fully invested, equally weighted top-decile winner strategy. **VM** scales MOM toward 12% annualized volatility using trailing 126-session shadow RMS risk. **FIX** holds VM's mean development-period target exposure. **FIXVOL** uses a development-calibrated constant exposure designed to match VM's overall volatility without dynamic timing.

From paper idea to an executable Indian-equity strategy

The original paper studies a leveraged winners-minus-losers portfolio. This implementation keeps the volatility-management mechanism while testing long-only NIFTY 500 winners plus cash. This is an adaptation of the paper's risk overlay, not a replication of its long-short portfolio.



Signal and eligibility

- At each month-end formation, compound 11 monthly total returns from t-12 through t-2. The most recent month is skipped.
- Start from the complete official membership roster. A missing panel observation remains visible and fails eligibility rather than silently shrinking the universe.
- Require dated identity and accepted EQ/BE/BZ classification, a valid formation quote, positive lagged 20-session ADV, and at least 15 positive-volume sessions in the preceding 20 sessions.
- Rank eligible names by momentum, break exact ties by permanent security ID, and select ceil(10% of the eligible roster).

Frozen portfolio variants

Account	Target equity allocation
MOM	100%
VM	Monthly 12% / risk, capped at 100%
FIX	Mean valid VM development target
FIXVOL	Fixed volatility-matched allocation

FIX and FIXVOL were calibrated only on development data. They use the same winner sleeve and account engine, isolating whether VM's time variation added value beyond a constant equity/cash mix.

Risk estimator and allocation rule

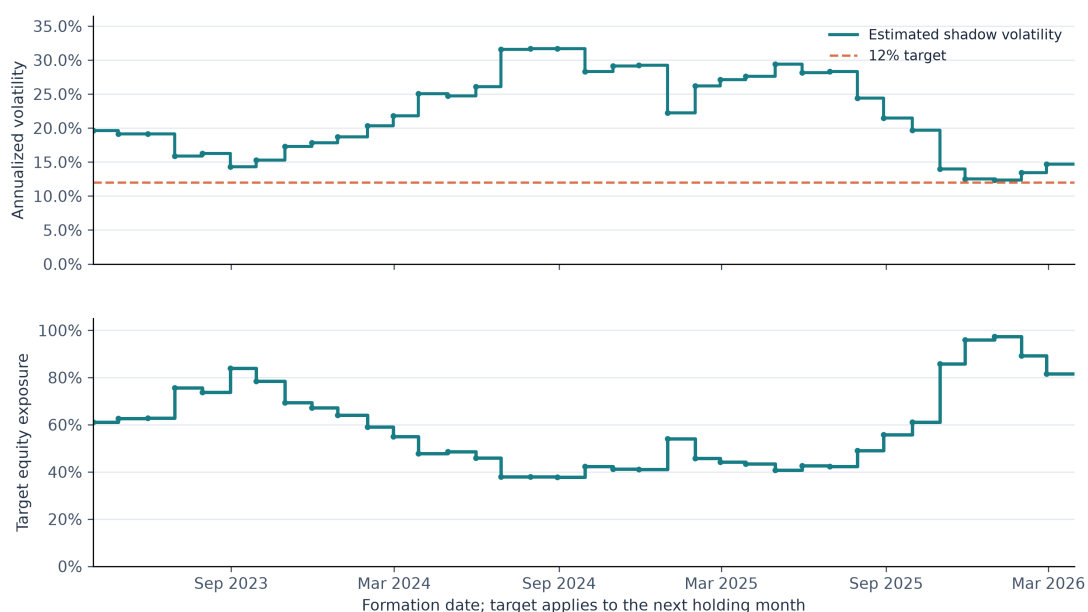
$\sigma_{\text{hat}}(d) = \sqrt{(252 / 126) \times \text{sum}(\text{last 126 shadow daily returns squared})}$

VM target equity = $\min(1, 0.12 / \sigma_{\text{hat}})$. This is an annualized second-moment or RMS estimate, not a demeaned rolling standard deviation. Missing, nonfinite or nonpositive estimates target cash with an explicit error status.

Holdout behavior of the frozen overlay

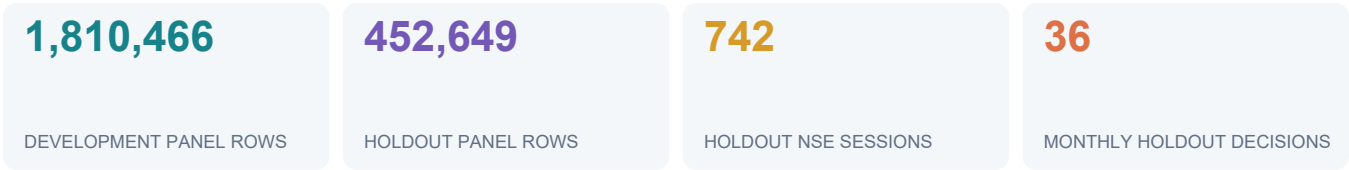
Volatility signal and resulting equity exposure

Frozen 126-session RMS estimate; monthly formation decisions | Holdout: Apr 2023-Mar 2026



The research contribution is the implementation discipline

Most backtest failures in this setting come from universe history, identity, corporate actions, liquidity or accounting rather than the ranking formula. The pipeline makes those choices explicit and auditable.



Point-in-time data

- Official August 2014 CNX 500 snapshot plus effective-dated official additions, deletions, replacements and later snapshots; interval rule `valid_from <= date < valid_to`.
- Dated ISIN and symbol lineage identify continuing securities without using identity evidence to infer index membership.
- Development membership counts remain 500-501, with zero unresolved identities, impossible prelisting members, alias collisions or duplicated economic securities.
- No missing return or volume is replaced by zero; no future observation repairs history; lagged ADV excludes today's observation.

Executable account mechanics

- Trade at the next eligible session's raw open using deterministic whole-share, cash-funded orders. Sells precede buys, but unsettled proceeds cannot finance purchases.
- Per-security daily notional is capped at 1% of preceding 20-session rupee ADV. Missing ADV creates no capacity. Rebalance orders expire after five NSE sessions.
- Fees plus square-root market impact, T+2 settlement-business-day cash availability, FIFO lots and the declared capital-gains tax scenario are maintained in one reconciled ledger.
- Positions, orders, fills, receivables, gains, tax liabilities and daily NAV reconcile; maximum reported NAV reconciliation error is effectively zero.

Corporate actions: preserve economics, never invent value

Class	Frozen holder treatment
Cash dividend	Gross declared cash per pre-event ordinary share; tax belongs in the account layer.
Split / ordinary bonus	Transform quantity and lot basis with verified multiplier.
Optional buyback	Passive holder does not participate; no direct adjustment.
Rights issue	No subscription capital; separately track and sell only with actual executable RE data, otherwise lapse.
Mandatory received security	Carry explicitly while unpriced; recognize from first accepted observable quote.

Holdout integrity.

Holdout-specific data and corporate-action issues were resolved from contemporaneous primary evidence using predeclared treatment rules; no signal, ranking, sizing, cost, tax, or execution parameter was changed in response to holdout performance.

Out-of-sample results: protection was real, timing alpha was not

The four accounts were run together from frozen 31 March 2023 states through 30 March 2026. Performance was not inspected during event resolution. The NIFTY 500 TRI is an untaxed published-index comparator, not an executable after-tax account.

Cumulative wealth: long-only momentum and risk controls

Executable post-cost/post-tax accounts; NIFTY 500 TRI benchmark | Holdout: 3 Apr 2023–30 Mar 2026

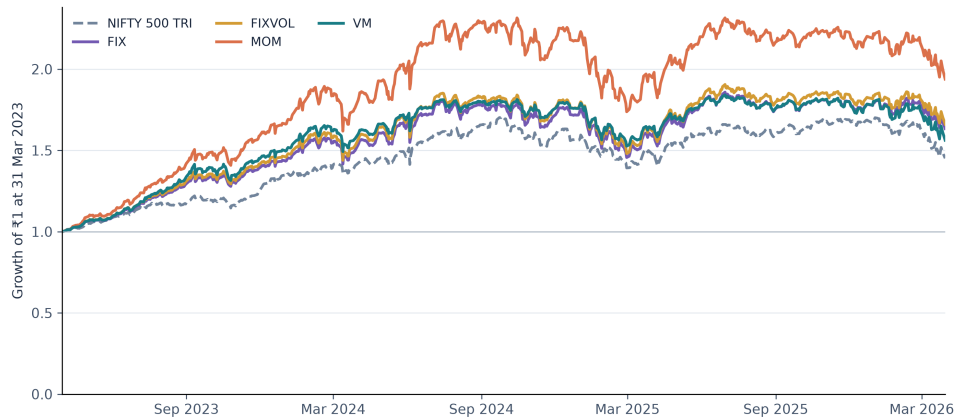


Figure 1. Holdout-only normalized wealth. MOM accumulated the most wealth; VM finished below both fixed-allocation controls. Every path begins at 1 because the four accounts have different frozen opening NAVs.

Performance interpretation

- **MOM:** strongest CAGR and Sharpe, with the deepest drawdown, highest volatility and highest turnover.
- **VM:** cut volatility by 5.97 percentage points and maximum drawdown by 9.13 points versus MOM, but surrendered 8.66 points of CAGR.
- **FIX / FIXVOL:** both exceeded VM's CAGR and Sharpe while taking moderately more risk, weakening the case that monthly exposure timing added value.
- All four accounts beat TRI CAGR, but the comparator is frictionless and untaxed. The spread is descriptive, not after-tax alpha.

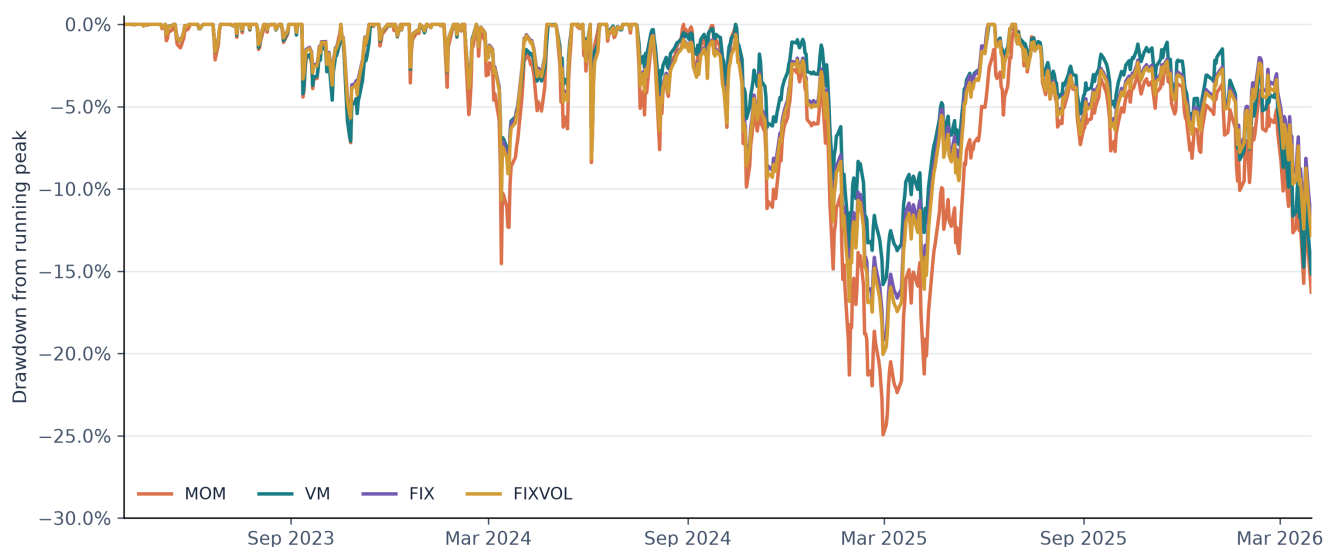
Development context

Account	CAGR	Vol	Sharpe	Max DD
MOM	8.86%	20.30%	0.53	-38.32%
VM	5.05%	14.23%	0.42	-34.42%
FIX	6.81%	13.50%	0.56	-27.00%
FIXVOL	7.11%	14.19%	0.56	-28.31%

Development covers April 2015–March 2023. FIX and FIXVOL are in-sample controls there because their constants use full-development information; the holdout comparison is the decisive test.

Holdout drawdowns

Executable post-cost/post-tax accounts | 3 Apr 2023–30 Mar 2026



Implementation frictions are material; conclusions remain bounded

Holdout implementation frictions

CAGR differences on matched account paths; the two bars are not additive

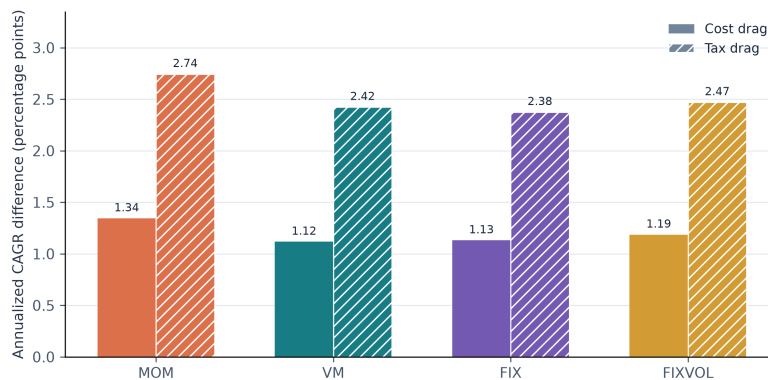


Figure 2. Annualized CAGR differences on matched executable account paths. Cost and tax bars are not additive contributions and are not independent counterfactual reruns.

Account	Gross turnover	Cost drag	Tax drag	Tax paid
MOM	5.34x	1.34 pp	2.74 pp	INR 2.34m
VM	4.73x	1.12 pp	2.42 pp	INR 1.36m
FIX	4.92x	1.13 pp	2.38 pp	INR 1.33m
FIXVOL	5.12x	1.19 pp	2.47 pp	INR 1.45m

Material limitations

- One immaterial UPL fractional bonus claim remained unpriced and was excluded from NAV rather than assigned an invented value.

Final assessment

What worked: the overlay reduced realized risk and drawdown exactly in the direction predicted by its design. The data and account implementation are explicit, auditable and reproducible.

What did not: VM did not beat simpler constant-exposure controls on holdout CAGR or Sharpe. The strongest claim is therefore risk control, not return enhancement.

Investment implication: use volatility scaling when drawdown containment and capital preservation have explicit value. Do not assume the changing exposure adds alpha without a stronger benchmark against fixed-risk allocations.

Sources and reproducibility

Barroso, P. and Santa-Clara, P., *Momentum Has Its Moments*, Journal of Financial Economics 116 (2015), 111-120. Project sources: official NSE membership notices and market data; accepted dated identity and corporate-action evidence; [results/evaluation_holdout/summary_metrics.csv](#); final freeze manifest; figure validation notes. AI coding assistance supported implementation, audit, documentation and figures.